

GlowBar — Product Innovation

GlowBar Zeste Citron

Snack Concept Summary

Document Type

Snack Concept Summary

Brand

GlowBar

Product Reference

GB-ZC-2026

Date

June 2026

Format

40 g bar — Soft & Crunchy

Status

Concept Stage

Segment

Premium Functional Snack

Dietary

100 % Plant-Based, Non-GMO

1. Product Positioning

GlowBar targets active, health-conscious consumers seeking a plant-based energy snack that does not compromise on taste or texture. The bar sits at the intersection of indulgence and nutrition — a bright citrus flavour, a satisfying dual texture (soft & crunchy), and a clean, fruit-forward ingredient list that invites confidence at the point of purchase.

GlowBar Zeste Citron is positioned in the **premium functional snack segment**, competing on ingredient quality and sensory experience rather than on price, targeting retail channels where wellness-oriented shoppers seek transparency, provenance, and genuine flavour authenticity.

2. Commercial Description

Packed with the sun-bright energy of real lemon zest, **GlowBar Zeste Citron** is the snack that refuses to choose between taste and nutrition. Every bite delivers a satisfying contrast of **moelleux + craquant** — a soft, chewy heart of dates and oats giving way to the satisfying crunch of puffed grains and seeds. 100 % plant-based and free from artificial additives, it is powered entirely by fruit-first ingredients you can actually pronounce.

Whether you need a natural energy boost before a workout, a mid-afternoon pick-me-up, or simply a snack that feels as good as it tastes — GlowBar Zeste Citron delivers pure citrus vitality in every 40 g bar. **Fruit-first. Plant-powered. Deliciously bright.**

Key Claims

Fruit-first recipe | 100 % plant-based | Moelleux + craquant texture | Natural energy boost | No artificial additives | Non-GMO certified

3. Motivation & Concept Rationale

GlowBar Zeste Citron is built on four interconnected pillars, each addressing a distinct consumer expectation in the premium functional snack category.

- **Fruit-First Philosophy**

The recipe leads with real fruit — lemon zest and fruit-based binders — rather than refined sugars or flavour concentrates. The goal is transparency on the ingredient panel and genuine flavour authenticity. Consumers can trust that the citrus character they taste reflects the citrus character they read on the label.

- **Moelleux + Craquant (Soft & Crunchy) Texture**

Consumers consistently rank texture among the top drivers of snack satisfaction. GlowBar Zeste Citron delivers a moist, chewy interior (from dates, oats, and fruit purée) paired with a crunchy coating or inclusions (puffed grains, seeds), creating a multi-sensory eating experience that drives repeat purchase and positive word of mouth.

- **100 % Plant-Based**

Every ingredient is of plant origin. No dairy, no gelatin, no animal-derived additives. This aligns with growing flexitarian and vegan consumer trends while broadening the addressable market to include label-conscious shoppers who may not identify as vegan but actively seek cleaner product profiles.

● Energetic & Functional

The macronutrient profile targets sustained energy: complex carbohydrates from oats and dates, healthy fats from seeds (sunflower, chia), and natural citrus flavour that conveys freshness and vitality. No added caffeine or synthetic stimulants — the energy proposition is clean, credible, and communicable.

4. Industrial-Scale Recipe Formulation

The following formulation is expressed on a **100 kg batch basis** and is intended as an indicative starting point for pilot-scale trials. All percentages are weight-for-weight (% w/w).

Ingredient	% of Total Batch (w/w)	Role / Function
Rolled oats (gluten-free certified)	30.0 %	Base structure, fibre, slow-release carbohydrates
Medjool date paste	22.0 %	Natural sweetener, binder, moisture retention
Puffed brown rice	10.0 %	Crunch texture, lightness, volume
Sunflower seed butter	8.0 %	Fat source, binding agent, creaminess
Lemon zest (fresh, dried)	4.0 %	Primary flavour — authentic citrus character

Ingredient	% of Total Batch (w/w)	Role / Function
Chia seeds	4.0 %	Omega-3 source, texture dots, nutrition claim support
Pumpkin seeds (pepitas)	5.0 %	Protein contribution, crunchy inclusions
Rice malt syrup	5.0 %	Binding agent, surface sheen, mild sweetness
Coconut oil (refined)	3.5 %	Fat carrier, mouthfeel improvement, shelf-life extension
Lemon juice concentrate	3.0 %	Acidity, flavour brightness and freshness
Natural lemon flavour (plant-derived)	0.8 %	Flavour top-note amplification and consistency
Sea salt	0.3 %	Flavour enhancer, balances sweetness
Sunflower lecithin	0.2 %	Emulsifier, texture homogeneity
Turmeric powder	0.2 %	Natural yellow colour, anti-inflammatory positioning
TOTAL	100.0 %	100 kg batch yield (before processing losses)

Formulation Notes

All ingredients are **non-GMO certified** and free from the 14 major allergens, except for sesame (from tahini-adjacent seeds) — **allergen advisory labelling is recommended**. Shelf-life target: **9 months** at ambient temperature. Final formulation to be confirmed following pilot trials and stability testing.

5. Industrial Manufacturing Process

The process below reflects an **industrial continuous-production model** with approximate parameters for temperature, time, and equipment settings. All steps are designed to be scalable to a commercial line output of approximately 500–800 kg/h.

Step 1 Ingredient Preparation & Dosing

Weigh all dry and wet ingredients separately using calibrated load cells to ± 0.1 % accuracy. Date paste is warmed to **40 °C** to reduce viscosity for consistent mixing. Lemon zest and juice concentrate are blended together and held at **4 °C** in refrigerated holding tanks prior to incorporation. Dry ingredients are pre-sieved to remove foreign matter.

Step 2 Dry Blend

Rollled oats, puffed brown rice, chia seeds, pumpkin seeds, and sea salt are combined in a **ribbon blender** (low speed, **3 min**) to achieve a uniform dry mix. Turmeric powder is added at this stage to ensure even colour distribution throughout the matrix before any wet ingredients are introduced.

Step 3 Wet Mix

Date paste, sunflower seed butter, rice malt syrup, and refined coconut oil are combined in a **heated jacketed mixer at 45 °C**. The lemon zest blend, lemon juice concentrate, and natural lemon flavour are incorporated. Sunflower lecithin is added last and mixed until fully emulsified (**5 min, medium speed**). Target viscosity: 18,000–22,000 cP at 45 °C.

Step 4 Combining (Wet + Dry)

The wet mixture is added progressively to the dry blend in the ribbon blender. Mixed at **low-to-medium speed for 4–6 minutes** until a cohesive, slightly sticky mass is obtained. Target moisture content: **14–16 %**. Overmixing must be avoided to preserve the structural integrity of the puffed rice inclusions.

Step 5 Sheeting & Cutting

The bar mass is fed into a **slab former / roller compaction unit**. Compressed to a target thickness of **12 mm** and density of approximately **1.05 g/cm³**. A guillotine cutter portions bars to

40 g ± 1 g (approx. 90 mm × 30 mm footprint). Off-cut trim is recycled upstream within the same shift.

Step 6 Optional Coating / Enrobing

Bars pass through a light **enrobing tunnel** with a thin layer of tempered rice-based chocolate or a lemon-flavoured yoghurt-style (plant-based) coating to reinforce the *craquant* (crunch) outer layer. Tunnel set at **28 °C**; cooling belt at **14 °C for 3 min** to set coating. Coating add-on: approximately 5–8 % of bar weight.

Step 7 Quality Control Checks

In-line **weight check** (every bar via checkweigher), **metal detection** (Fe ≤1.5 mm / SUS ≤2.0 mm), and periodic manual texture, moisture (target **a_w < 0.65**), and Brix checks. Non-conforming bars are rejected and recycled upstream within the same production shift. Rejection rate KPI: < 1.5 %.

Step 8 Packaging & Labelling

Bars are individually **flow-wrapped** in compostable cellulose-based film (moisture vapour transmission rate < 5 g/m²/day). Secondary packaging: recyclable cardboard display box (12-bar count). Lot code and best-before date are printed **inline** by thermal inkjet coder. Storage and transport conditions: ambient, < **25 °C**, away from direct light and strong odours.

6. Nutritional Overview

Estimated nutritional values per **40 g serving** (one bar), based on recipe formulation. All values are approximate pending laboratory validation.

Nutrient	Per 40 g Bar
Energy	~170 kcal / 710 kJ
Carbohydrates	~24 g

Nutrient	Per 40 g Bar
<i>of which sugars</i>	<i>~12 g (all natural)</i>
Fat	~6 g
<i>of which saturates</i>	<i>~1.5 g</i>
Protein	~4 g
Dietary Fibre	~3.5 g
Sodium	~55 mg

Analytical Note

Values are estimates based on recipe formulation using standard food composition databases. **Final nutritional data must be validated by accredited laboratory analysis** prior to label printing and product launch. Front-of-pack labelling to comply with applicable regional regulation.

7. At-a-Glance Comparison Summary

The table below is formatted for rapid side-by-side comparison with other snack concepts under review. All attributes reflect the current concept-stage specification.

▼ GlowBar Zeste Citron — Quick Reference Card

Attribute

GlowBar Zeste Citron

Format

Soft & crunchy bar, 40 g individual portion

Flavour Profile

Lemon zest, citrus-forward, fruit-bright

Texture Profile

Moelleux + craquant (chewy interior / crunchy coat & inclusions)

Dietary Status

100 % plant-based, non-GMO certified

Key Ingredients

Rolled oats, Medjool dates, lemon zest, pumpkin seeds, chia seeds

Allergens

Free from 14 major allergens; sesame advisory recommended

Energy (per bar)

~170 kcal / 710 kJ

Target Consumer

Active adults, flexitarians, wellness-focused shoppers

Shelf Life

9 months at ambient temperature (< 25 °C)

Packaging

Compostable individual flow-wrap; recyclable cardboard secondary box (12-bar)

Market Positioning

Premium functional snack segment

Key Differentiators

Fruit-first recipe, dual texture, clean label, compostable packaging

Development Stage

Concept — pending pilot trial and lab validation